Mastering Spring AI

The Java Developer's Guide for Large Language Models and Generative AI

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To my beloved wife, Vijaya Anirudhran, whose unwavering support and love have been my anchor through every journey. And to my wonderful children, Pooja Anirudhran and Deepika Anirudhran, who inspire me every day with their curiosity, resilience, and boundless potential. This book is a testament to the love and strength of our family.

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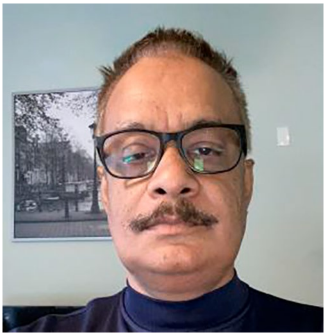
About the Author



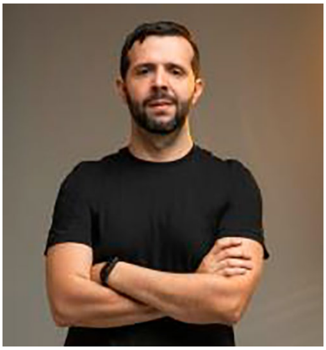
Banu Parasuraman is a seasoned technologist with over 30 years of experience in the IT industry, currently focusing on the transformative fields of AI and automation. Throughout his distinguished career, Banu has provided invaluable advisory services to clients across a multitude of industries, guiding them in leveraging AI-driven solutions and automation technologies to enhance their operations and achieve digital transformation. His expertise has been sought by over 25 leading companies spanning sectors like retail, healthcare, logistics, banking, manufacturing, automotive, oil and gas, pharmaceuticals, media, and entertainment across the United States, Europe, and Asia.

A passionate advocate for AI and automation, Banu has been at the forefront of encouraging organizations to adopt these technologies early, helping them to streamline operations, improve decision-making, and maintain a competitive edge. His deep knowledge extends across all major AI and automation platforms, including advanced cloud services and tools that integrate seamlessly into enterprise environments.

Banu is a respected speaker, having participated in numerous external engagements such as VMworld, SpringOne, Spring Days, and Spring Developer Forum Meetups, where he shared his insights with CXOs and engineers alike. His internal engagements include leading workshops on AI-driven solutions and automation strategies, where he has been instrumental in enabling sales plays and strategies focused on these cutting-edge technologies.

Beyond speaking engagements, Banu is an active thought leader in the industry, contributing to the wider adoption of AI and automation through his blogs on Medium and LinkedIn, where he shares his knowledge and promotes best practices in AI and automation development. He is also the author of the book *Practical Spring Cloud Function: Developing Cloud-Native Functions for Multi-Cloud and Hybrid-Cloud Environments*, available on Amazon. Banu's work continues to shape the future of technology, inspiring organizations to innovate and embrace the power of AI and automation.

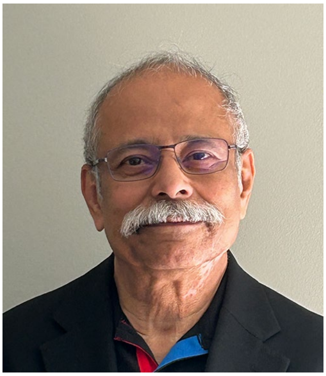
About the Technical Reviewers



Andres Sacco has been a professional developer since 2007, working with various languages, including Java, Scala, PHP, Node.js, and Kotlin. His background is mainly in Java and its associated libraries or frameworks, like Spring, JSF, iBATIS, Hibernate, and Spring Data. He is focused on researching new technologies to improve the performance, stability, and quality of the applications he develops.

In 2017, he started to find new ways to optimize data transference between applications to reduce the cost of infrastructure. He suggested some actions, some applicable to all the microservices and others to just a few of them; as a result, the cost was reduced by 55%. Some of these actions are connected directly with the bad use of the databases.

He recently published some books with Apress about the last version of Scala and Spring Data. He also published a set of theoretical-practical projects on uncommon testing methods, such as architecture tests and chaos engineering.



Debasish Banerjee, Ph.D., is a seasoned thought leader, hands-on architect, and practitioner of cutting-edge technologies. He has a proven track record spanning more than a decade of advising and closely working with Fortune 500 and other customers in the United States, Europe, and Asia. He recently joined Guild Systems Inc (`www.guildsystems.com`) as the ITOps and FinOps lead. Until September 2024, he was a principal Customer Success Manager at IBM. Debasish's work successfully established several advanced IBM technologies in the field and has generated several hundred million USD in revenue for IBM.

Debasish's present areas of interest are superior observability to reduce the mean time to recovery, automatic application resource management, FinOps, and the use of AI to enhance these technologies. Debasish obtained his Ph.D. in combinator-based functional programming languages. He is the father of two brilliant software engineer daughters, Cheenar and Neehar, who are his pride and joy.

Acknowledgments

I would like to express my deepest gratitude to the individuals who played a crucial role in bringing this book to life.

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Finally, I would like to acknowledge the entire **Apress staff** who contributed to the creation of this book. Your professionalism, support, and collaborative efforts were essential in bringing this project to fruition. Thank you for your hard work and commitment to excellence.

This book is a result of the collective efforts of all these remarkable individuals, and I am truly thankful for their contributions.

CHAPTER 1

Introduction to Generative AI and Large Language Models (LLMs)

In the constantly evolving realm of artificial intelligence (AI) and machine learning (ML), few advancements have been as revolutionary in recent years as the ascent of Generative AI and Large Language Models (LLMs). These cutting-edge models have redefined our comprehension of AI systems, unleashing a wave of unprecedented opportunities and applications spanning diverse fields. In this chapter, we will embark on an enlightening exploration of the underpinnings, evolution, and profound impact of Generative AI and Large Language Models. Moreover, we will examine how Spring AI seamlessly integrates into this dynamic landscape, shaping the future of AI-driven solutions.

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1.1 Understanding the Basics of Artificial Intelligence

Definition and Historical Perspective of AI

Artificial intelligence, often abbreviated as AI, is a multidisciplinary field of computer science that aims to create systems and machines capable of performing tasks that typically require human intelligence. These tasks encompass a wide range of activities, from problem-solving and decision-making to natural language understanding and perception.

The concept of AI has been a part of human imagination for centuries, with early myths and stories featuring mechanical beings capable of human-like thought. However, it wasn't until the mid-20th century that AI as we know it today began to take shape.

The term "artificial intelligence" was coined by John McCarthy in 1956 during the Dartmouth Workshop, which marked the birth of AI as a formal academic discipline. Over the decades, AI has undergone several waves of enthusiasm and disappointment, known as AI winters, where the expectations often exceeded the capabilities of the technology at the time.

Today, AI has become an integral part of our lives, powering intelligent virtual assistants, recommendation systems, autonomous vehicles, and much more. Its historical journey from myth to reality highlights the persistence and determination of researchers and engineers to create machines that can mimic human intelligence. See Figure 1-1.

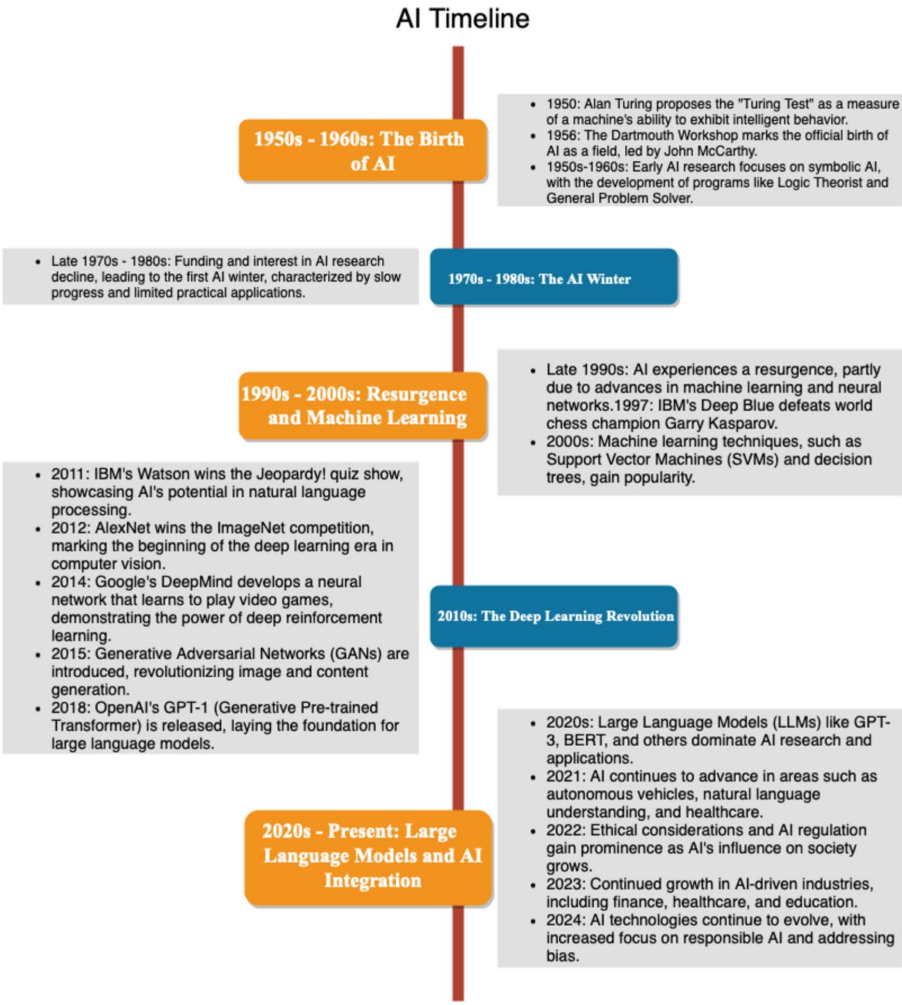

Figure 1-1. *AI timeline*

Key Concepts in AI: Algorithms, Neural Networks, and Data Processing

To understand AI fully, it's essential to grasp some key concepts that underpin the field. These concepts include algorithms, neural networks, and data processing.

Algorithms: Algorithms are step-by-step sets of instructions designed to solve specific problems or perform particular tasks. In AI, algorithms are crucial for decision-making and problem-solving. They are the building blocks of AI systems, guiding how data is processed, analyzed, and acted upon.

Neural Networks: Neural networks are computational models inspired by the structure and function of the human brain. They consist of interconnected nodes, or "neurons," organized in layers. Neural networks are capable of learning patterns from data through a process called training, making them central to many AI applications, including image recognition, natural language processing, and speech recognition.

Data Processing: Data is the lifeblood of AI. Machine learning algorithms[1] rely on large datasets to learn and make predictions or decisions. Data processing involves collecting, cleaning, and transforming data into a format suitable for AI models. The quality and quantity of data significantly impact AI performance.

[1] Algorithms are automated instructions that can vary in complexity based on their depth. Machine learning and artificial intelligence are both based on sets of algorithms, with the key difference being how they handle structured vs. unstructured data

The Role of AI in Modern Technology and Society

AI's role in modern technology and society is profound and far-reaching. It has permeated various aspects of our lives, from the devices we use to the services we access. Here are some key areas where AI plays a pivotal role:

1. **Healthcare**: AI is revolutionizing healthcare by enabling faster and more accurate diagnosis, drug discovery, and personalized treatment plans. Machine learning models can analyze medical images, detect anomalies, and predict patient outcomes.

2. **Finance**: In the financial industry, AI is used for fraud detection, algorithmic trading, and risk assessment. AI-driven chatbots and virtual assistants also enhance customer service and support.

3. **Education:** AI-powered educational tools offer personalized learning experiences, adapt to individual student needs, and provide valuable insights to educators. This promotes more effective learning outcomes.

4. **Natural Language Processing**: AI models excel in understanding and generating human language. This capability drives advancements in virtual assistants like Siri and Alexa, as well as language translation services.

5. **Autonomous Vehicles**: Self-driving cars rely on AI algorithms to navigate and make split-second decisions based on sensor data. This technology has the potential to reshape transportation systems.

6. **Entertainment**: AI is used in content recommendation systems, video game design, and even the creation of art and music.

As AI continues to advance, it poses both exciting opportunities and ethical challenges. It is essential to strike a balance between innovation and responsible use to harness the full potential of AI for the benefit of society.

1.2 The Journey from Traditional Machine Learning to Generative AI

Fundamentals of Machine Learning: Supervised, Unsupervised, and Reinforcement Learning

Machine learning (ML) serves as the foundation upon which Generative AI and Large Language Models are built. ML is a subset of AI that focuses on developing algorithms capable of learning from data and making predictions or decisions without explicit programming.

There are several categories of machine learning, with three primary paradigms being supervised learning, unsupervised learning, and reinforcement learning:

Supervised Learning: In supervised learning, algorithms are trained on labeled datasets, where the correct answers are provided. The goal is to learn a mapping from inputs to outputs, allowing the model to accurately predict unseen data. This paradigm is used for tasks like classification and regression.

Unsupervised Learning: Unsupervised learning deals with unlabeled data and aims to discover patterns, structures, or groupings within the data. Common techniques include clustering and dimensionality reduction. Unsupervised learning can uncover hidden insights and relationships in data.

Reinforcement Learning: Reinforcement learning is centered around agents that interact with an environment and learn to make decisions by receiving feedback in the form of rewards or penalties. It is commonly used in robotics, game playing, and autonomous systems.

These fundamental ML paradigms laid the groundwork for the transition to more complex and capable AI systems.

Transition to Deep Learning: Importance and Impact

Deep learning, a subfield of machine learning, emerged as a game-changer in the world of AI. At its core, deep learning leverages neural networks with multiple layers, often referred to as deep neural networks. This architecture enables models to automatically learn hierarchical features from data, making them capable of handling tasks that were previously challenging for traditional ML algorithms.

The importance of deep learning lies in its ability to extract high-level representations from raw data. This feature is particularly valuable for tasks such as image recognition, speech recognition, and natural language processing. Convolutional neural networks (CNNs) and recurrent neural networks (RNNs) are examples of deep learning architectures that have revolutionized computer vision and sequential data processing.

The impact of deep learning can be observed in applications like autonomous vehicles, where CNNs analyze images from cameras mounted on vehicles to make real-time driving decisions. Additionally, deep learning has fueled breakthroughs in speech recognition technology, enabling virtual assistants to understand and respond to spoken commands accurately.

Emergence of Generative AI: Characteristics and Capabilities

Generative AI represents a significant leap in AI capabilities. Unlike traditional AI systems that focus on specific tasks, Generative AI aims to create new content, whether it's text, images, or other forms of data. These systems have the remarkable ability to generate human-like content, making them a cornerstone of natural language generation and creative applications.

The emergence of Generative AI is marked by the development of Large Language Models (LLMs), which have set new standards for AI performance. LLMs are neural network–based models with a vast number of parameters, enabling them to understand and generate human language at a level

1.3 Exploring Large Language Models: A Paradigm Shift in AI

What Are Large Language Models (LLMs)?

Large Language Models (LLMs) are a type of deep learning model that has gained immense popularity and attention in recent years. These models are characterized by their enormous size, often comprising hundreds of millions or even billions of parameters. LLMs are specifically designed for natural language understanding and generation tasks.

The key defining feature of LLMs is their ability to process and generate human language and interpret and generate images. They can understand the context of text, answer questions, translate languages, summarize documents, and even create creative written content. Some of the models like Dall-E can generate images. This paradigm shift in AI has led to advancements in fields such as chatbots, language translation, content generation, and more.

Understanding the Mechanism: How LLMs Process and Generate Language

The mechanism behind Large Language Models is rooted in deep learning and neural networks. These models utilize a variant of neural networks known as Transformer architectures. The Transformer architecture introduced the concept of self-attention mechanisms, allowing the model to weigh the importance of different words in a sentence when making predictions. This was articulated in the paper "Attention Is All You Need" https://arxiv.org/abs/1706.03762.

The training process of LLMs involves exposing the model to vast amounts of text data. Through this exposure, the model learns the statistical patterns, grammar, semantics, and context of human language. This knowledge is encoded in the model's parameters, which are fine-tuned during training.

When generating language, LLMs take an input sequence of text and use their learned knowledge to predict the next word or sequence of words. This process is repeated iteratively, resulting in coherent and contextually relevant text generation. LLMs can be used for various tasks, from completing sentences to generating entire articles.

Comparing LLMs with Traditional Machine Learning Models

To appreciate the significance of LLMs, it's essential to compare them with traditional machine learning models, especially in the context of natural language processing (NLP) tasks.

> **Traditional NLP Models**: Traditional NLP models often rely on handcrafted features and linguistic rules. They require substantial manual effort for feature engineering, making them less adaptable to different languages and domains. These models may struggle with understanding context and generating coherent text.

Large Language Models (LLMs): LLMs, on the other hand, excel at capturing complex language patterns without the need for extensive feature engineering. They are pretrained on massive text corpora, enabling them to generalize across languages and domains. LLMs are highly versatile and can be fine-tuned for specific tasks with relatively little data.

The comparison between LLMs and traditional NLP models underscores the transformative impact of LLMs on AI and language-related applications. Their ability to process and generate language with human-like fluency has opened new horizons for AI-powered communication and content creation. Figure 1-2 shows the comparison between traditional machine learning and LLMs.

Feature	Traditional Machine Learning	Large Language Models (LLMs)
Data Requirements	Generally, requires less data. Models are often trained on specific datasets tailored to a particular task.	Requires massive amounts of data, often sourced from diverse and extensive text corpora to learn language patterns.
Training Complexity	Relatively simpler models that can be trained on modest hardware depending on the complexity and size of the dataset.	Requires significant computational resources (often GPUs or TPUs) due to the complexity and size (billions of parameters).
Task Specialization	Models are usually task-specific and need to be retrained or finely tuned for each new task.	Designed to handle multiple tasks with a single model, often without task-specific tuning (zero-shot or few-shot learning).
Interpretability	Often more interpretable, especially with models like decision trees and linear regression.	Less interpretable due to their complexity and the 'black box' nature of deep neural networks.
Generalization	Depends on the model and the feature engineering; some models generalize better than others.	Excellent at generalizing from the training data to a wide range of natural language tasks due to extensive pre-training.
Performance on New Tasks	Requires retraining or adaptation to handle new tasks effectively.	Can perform new language tasks they were not explicitly trained on, demonstrating adaptability and flexibility.
Deployment	Often lighter and easier to deploy on limited resources depending on the model.	Due to their size, deployment can be resource-intensive and may require specialized infrastructure like cloud services.
Development and Maintenance Cost	Lower development costs and maintenance can be manageable depending on the application and model complexity.	High development cost due to data acquisition, computational resources, and ongoing training requirements.
Model Updates	Models may need periodic retraining and updates based on new data or changes in the data distribution.	Continuous updates may be necessary to incorporate new information, refine responses, and improve model performance.

Figure 1-2. *Traditional machine learning vs. LLMs*

Large Language Models (LLMs) are highly adaptable and can be extended with ease to perform a variety of tasks. Their ability to process and generate human-like text allows them to be fine-tuned or adapted for specific domains, making them versatile for numerous applications.

Extending LLMs typically involves adjusting them to handle new types of data, integrating them with other systems, or enhancing their capabilities through additional training or customization. This flexibility makes LLMs a powerful tool in AI-driven solutions.

1.4 Overview of Prominent LLMs: GPT, BERT, and Beyond

Large Language Models (LLMs) have revolutionized the field of artificial intelligence, especially in natural language processing (NLP) and generation tasks. These models are characterized by their vast size, measured in billions of parameters, and their ability to understand, generate, and interpret human language with remarkable accuracy. Let's take a look into some of the most prominent LLMs, highlighting their development, capabilities, and impact. See Figure 1-3.

1. **GPT Series (OpenAI) (`https://platform.openai.com/docs/models`)**

    - **Overview**: The Generative Pretrained Transformer (GPT) series by OpenAI is among the most well-known LLMs. Starting with GPT-1, the series has evolved significantly over time, with GPT-3, and its successors setting new standards for language model capabilities.

    - **Capabilities**: GPT models are pretrained on a diverse range of Internet text. They excel in tasks such as text generation, conversation, translation, and answering questions without specific fine-tuning for each task.

- **Impact**: GPT-3 and its successors have been widely adopted across industries for applications like content creation, coding assistance, and customer support, showcasing the versatile potential of LLMs in real-world applications.
- Notes[2] on GPT-4:
- **Release Date**: GPT-4's release in 2023 marked another significant leap in the capabilities of language models, showcasing OpenAI's continued innovation in AI.
- While the exact number of parameters for GPT-4 has not been disclosed, it is known to be substantially larger than GPT-3, indicating a further increase in complexity and understanding capabilities.
- **Capabilities and Applications**: GPT-4 advances the state of the art with more sophisticated text generation, improved context understanding, and enhanced performance across a broad range of language tasks. It has been applied to even more complex and nuanced applications than its predecessors, demonstrating the ongoing evolution of AI's potential to mimic and augment human intelligence in writing, conversation, and analysis tasks.

[2] At the time of this publication

2. **BERT (Google) (https://huggingface.co/docs/transformers/en/model_doc/bert)**

    - **Overview**: Bidirectional Encoder Representations from Transformers (BERT) is a breakthrough model introduced by Google. Unlike GPT, BERT focuses on understanding the context of words in sentences from both directions (left and right of a word), which is a departure from previous models that processed text in one direction.

    - **Capabilities**: BERT's bidirectional training improves its performance on a variety of language understanding tasks, including question answering, language inference, and named entity recognition.

    - **Impact**: BERT has been integrated into Google Search, enhancing the search engine's understanding of the nuances and context of user queries.

3. **T5 (Google) (https://huggingface.co/docs/transformers/en/model_doc/t5)**

    - **Overview**: Text-to-Text Transfer Transformer (T5) converts all NLP problems into a unified text-to-text format, where both input and output are always text strings. This approach simplifies the NLP pipeline and allows for a more generalized application of the model.

    - **Capabilities**: T5 has shown strong performance across a range of tasks, including translation, summarization, question answering, and classification tasks.

- **Impact**: T5's versatility and effectiveness have made it a popular choice for researchers and practitioners looking for a flexible and powerful model for various NLP tasks.

4. **ERNIE (Baidu) (`https://huggingface.co/docs/transformers/en/model_doc/ernie`)**

   - **Overview**: Enhanced Representation through Knowledge Integration (ERNIE) is a language model developed by Baidu. ERNIE is designed to better understand the semantics of language by integrating world knowledge and linguistic context into its pretraining.
   - **Capabilities**: ERNIE performs well on tasks requiring semantic understanding, such as sentiment analysis, named entity recognition, and question answering.
   - **Impact**: ERNIE's ability to incorporate external knowledge and context has helped improve the performance of language models on tasks requiring a deeper understanding of language and world knowledge.

5. **XLNet (Google/CMU) (`https://huggingface.co/docs/transformers/en/model_doc/xlnet`)**

   - **Overview**: XLNet is an advanced model that outperforms BERT on several benchmarks by using a permutation-based training strategy. This approach allows XLNet to learn the bidirectional context of a word more effectively.

- **Capabilities**: XLNet excels in tasks requiring understanding of complex sentence structures and has set new records in performance for tasks like text classification, question answering, and textual entailment.
- **Impact**: XLNet's innovative training strategy and its resulting performance gains have contributed to the ongoing research and development in the field, pushing the boundaries of what's possible with LLMs.

6. **LLaMA (Meta) (`https://llama.meta.com/docs/get-started/`)**
   - **Overview**: LLaMA, developed by Meta, is one of the latest advancements in the field of Large Language Models. It stands out for its efficiency and adaptability, designed to perform a wide range of language understanding and generation tasks with fewer parameters compared to its predecessors, making it more accessible for research and practical applications.
   - **Capabilities**: LLaMA excels in various NLP tasks such as text generation, translation, and complex reasoning. Its smaller model variants allow for use in environments with limited computational resources while maintaining high performance.
   - **Impact**: The versatility and efficiency of LLaMA make it suitable for both academic researchers and industry practitioners. It opens up new possibilities for deploying advanced NLP models in more diverse settings, including on devices with lower processing power.

7. **Mistral AI (`https://docs.mistral.ai/getting-started/models/`)**

    - **Overview**: Mistral AI represents a step forward in democratizing access to powerful language models. It aims to provide scalable solutions that are adaptable to a wide array of tasks, maintaining high performance without the extensive resource requirements typically associated with such models.

    - **Capabilities**: Mistral AI models are known for their general-purpose language understanding and generation, capable of performing tasks ranging from summarization and translation to more complex reasoning and content creation.

    - **Impact**: The adaptability and scalability of Mistral AI's models facilitate their integration into various applications, making advanced NLP capabilities accessible to a broader range of users and developers across industries.

8. **IBM Granite (`https://www.ibm.com/products/watsonx-ai/foundation-models?utm_content=SRCWW&p1=Search&p4=43700078747002614&p5=e&p9=58700008385933274&gbraid=0AAAAAD-_QsSKeLf0GDy8HjRfPsKdeUR-n&gclid=Cj0KCQjwz7C2BhDkARIsAA_SZKY4xgajHusdNGngUbre8IaTrSCJYp67GIjgcZBwrr8dxRH4XuoAdSYaAv6eEALw_wcB&gclsrc=aw.ds`)**